Step 8: The classification error rate difference for all OOB employee evaluation testing datasets are combined to analyze the importance of specific feature e_j. The average classification error rate difference for all OOB employee evaluation testing datasets provides the feature importance score. The importance of a specific employee evaluation feature (e_j) was computed by equation 11.10.

$$Importance_{e_j} = \frac{\sum_{j=1}^{v} ER_j''}{v} \tag{11.10}$$

where $Importance_{e_j}$ is the importance score for employee evaluation feature e_j. The greater the classification error rate difference, the higher the importance scores for the employee evaluation feature, and the more important the feature is. Employee evaluation features with higher importance scores are selected for assisting the further employee performance calculation.

The optimal employee evaluation feature vector is constructed with 'm' features having a higher importance score. The optimal employee evaluation dataset is constructed by projecting the dataset obtained from section 11.4.3 on the optimal employee evaluation feature vector and is defined by equation 11.11.

$$E_{n\times m} = E'_{n\times l} \times H \tag{11.11}$$

where $E_{n\times m}$ means the 'm' dimensional employee evaluation dataset constructed after feature selection, $E'_{n\times l}$ denotes the 'l' dimensional employee evaluation dataset constructed after feature extraction, and H represents the optimal evaluation feature vector.

11.3.5 Employee Performance Classification Using a Weighted Multilayer Support Vector Machine (WM-SVM)

The performance of employees is classified into high, moderate, and low categories based on employee evaluation data and an optimal evaluation feature vector using a WM-SVM. Assigning various weights to individual data points allows the WM-SVM training algorithm to learn the decision surface based on the relative relevance of data points in the employee assessment dataset.

Assignment of different weights to different data points and multinomial classification of employee performance differentiates the proposed WM-SVM from traditional SVM.

The training employee evaluation dataset for WM-SVM is defined by equation 11.12.

$$\left\{\left(e_j, y_j, G_j\right)\right\}_{j=1}^{m}, e_j \in E_{n\times m}, y_j \in \{-1,1\} \tag{11.12}$$

where G_j means the weight assigned to the employee evaluation data point, e_j represents the employee evaluation data point, y_j means the classification output, and $E_{n\times m}$ represents the employee evaluation dataset with 'n' employees and 'm' dimensions.

For the N-class classification problem, N (N-1)/2 classifiers are constructed based on the one-against-one method. Here the value of N is three because employee performance is classified into three classes. Each classifier is trained on samples from the two corresponding classes.

Consider a classifier S trained on samples from two classes a and b. The major goal of the WM-SVM is to create a hyperplane that differentiates employee evaluation data points in an N-dimensional space.

The separating hyperplane (i.e., the optimal margin hyperplane) must be constructed in such a way that maximizes the margin between the classes. The hyperplane margin is represented by $2/\| w^{ab} \|$. The optimization problem for WM-SVM is defined by equation 11.12.

$$\min_{y,k,\xi} \frac{1}{2}\left\|w^{ab}\right\|^2 + A\sum_j G_j^{ab}\xi_j^{ab} \tag{11.13}$$

$$\textit{Subject to } y_j\left((w^{ab})^2\varnothing\left(e_j^{ab}\right)+k^{ab}\right)\geq 1-\xi_j^{ab}$$

$$\xi_j^{ab}\geq 0,\ j=1,\ \ldots\ \ldots.\ m$$

where $\| w^{ab} \|/2$ denotes the inverse of hyperplane margin, ξ means the slack parameter, A depicts the regularization factor, k denotes the bias, $\varnothing$ is the mapping function that maps the training data to the classification workspace, and $A\sum_j G_j\xi_j$ represents the regularization term (introduced for reducing the training error).

The hyperplane margin can be maximized by minimizing equation 11.13. The employee performance decisions are performed based on equation 11.14.

$$\left((w^{ab})^2\varnothing\left(e_j^{ab}\right)+k^{ab}\right)=\left(\sum_{j=1}^{m} y_j\alpha_j^{ab}B\left(e_j^{ab},e^{ab}\right)+k^{ab}\right) \tag{11.14}$$

where α means the Lagrange constant and it must be ≥ 0, and $B(e_j, e)$ means the kernel function.

The sign $(w^2\varnothing(e_j)+k)$ determines whether the data sample is classified into a^{th} or b^{th} class. If employee performance data e_j^{ab} is classified into a^{th} class, then the vote for the a^{th} class of sample e_j^{ab} increased by one and if it is classified into b^{th} class, then the vote for the b^{th} class of sample e_j^{ab} increased by one.

After all the classifiers are trained, a voting strategy is used for decision-making. Then e_j^{ab} got classified into the class which gets maximum votes. The final output is in the range of –1 to 1.

Figure 11.4 depicts the employee performance classification in three performance classes. An employee is classified into the low-performance class if the output result (y_j) is $-1\leq y_j<0$, the moderate-performance class if the output result (y_j) is 0, and the high-performance class if the output result (y_j) is $0<y_j\leq 1$.

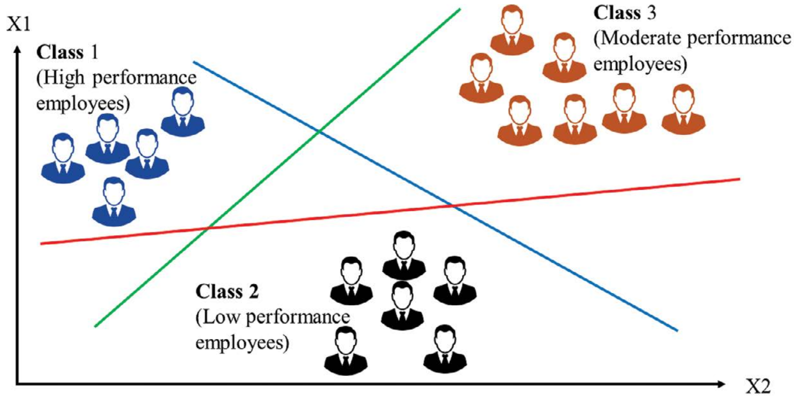

FIGURE 11.4 Employee classification based on their performance using a WM-SVM.

11.4 RESULTS AND DISCUSSION

This section deals with the analysis of the efficacy of the proposed employee performance evaluation model. The efficacy of the proposed WM-SVM–based employee performance evaluation model was compared to that of existing employee performance evaluation models, namely the Bayesian belief network model (BBN), fusion particle swarm optimization algorithm (FPSO), interval valued fuzzy weighted distance algorithm (IVFWDA), and classification and regression tree (C&RT).

The performance indicators used in this analysis are accuracy, precision, recall, classification error rate, and time consumption (Rani et al., 2022). Accuracy is described as the ratio of correctly classified employee evaluation samples to the total number of employee evaluation samples. It is computed using equation 11.15.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{11.15}$$

where TN means the number of employees classified accurately into high- or moderate-performance class, TP means the number of employees classified accurately into low-performance class, FP means the number of employees belonging to high- or moderate-performance class who are misclassified into low-performance class, and FN denotes the number of employees belonging to the low-performance class who are misclassified into the high- or moderate-performance class (Khang, Rani & Sivaraman, 2022).

Figure 11.5 illustrates the comparative investigation of employee performance classification accuracy of different performance evaluation models. The proposed WM-SVM–based employee performance evaluation model exhibited higher accuracy for classifying employees based on their performance into three classes compared to existing performance evaluation methods like BBN, FPSO, IVFWDA, and C&RT (Khang et al., 2022a).

This showed that the proposed model is efficient in evaluating the performance of employees in enterprises. In addition, accuracy for predicting high-performance

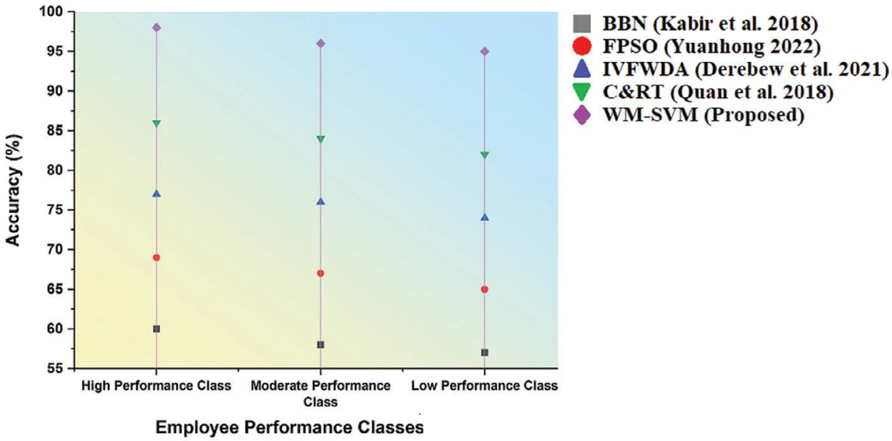

FIGURE 11.5 Comparison of employee performance classification accuracy of different models.

employees using WM-SVM was higher than that for predicting moderate- and low-performance employees using WM-SVM, as shown in Figure 11.5.

Precision is described as the percentage of employees who are classified accurately into low-performance class out of the total number of employees classified into low-performance class. It is defined by equation 11.16.

$$Precision = \frac{TP}{TP + FP} \tag{11.16}$$

Figure 11.6 illustrates the comparative investigation of employee performance classification precision of different performance evaluation models. The proposed

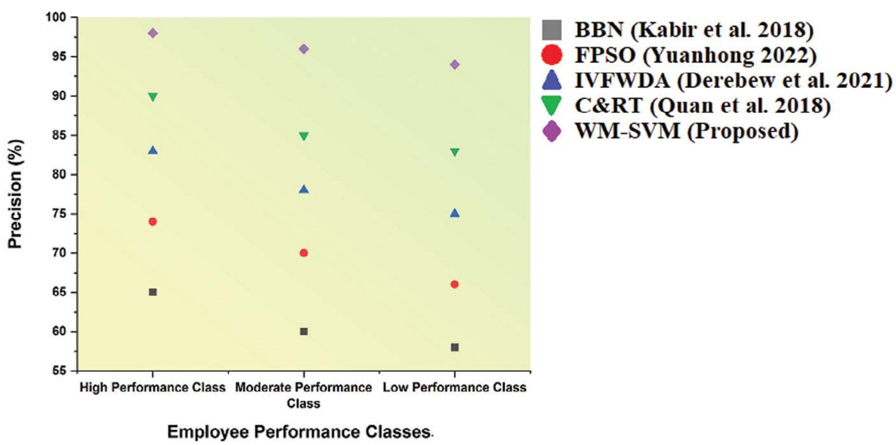

FIGURE 11.6 Comparison of employee performance classification precision of different models.

WM-SVM–based employee performance evaluation model exhibited higher precision for employee performance classification compared to existing performance evaluation methods like BBN, FPSO, IVFWDA, and C&RT.

The number of employees belonging to high- or moderate-performance classes who are misclassified into the low-performance class is minimal with the WM-SVM-based employee performance evaluation model compared to existing models (Hahanov et al., 2022b).

The recall is described as the percentage of employees who are classified accurately into the low-performance class out of the total number of employees belonging to the low-performance class. It is defined by equation 11.17.

$$Recall = \frac{TP}{TP + FN} \tag{11.17}$$

The proposed WM-SVM-based employee performance evaluation model exhibited higher recall for employee performance classification compared to existing performance evaluation methods such as BBN, FPSO, IVFWDA, and C&RT, as shown in Figure 11.6.

The number of employees belonging to the low-performance class who are misclassified into high- or moderate-performance classes is minimal with the WM-SVM-based employee performance evaluation model compared to existing models as shown in Figure 11.7.

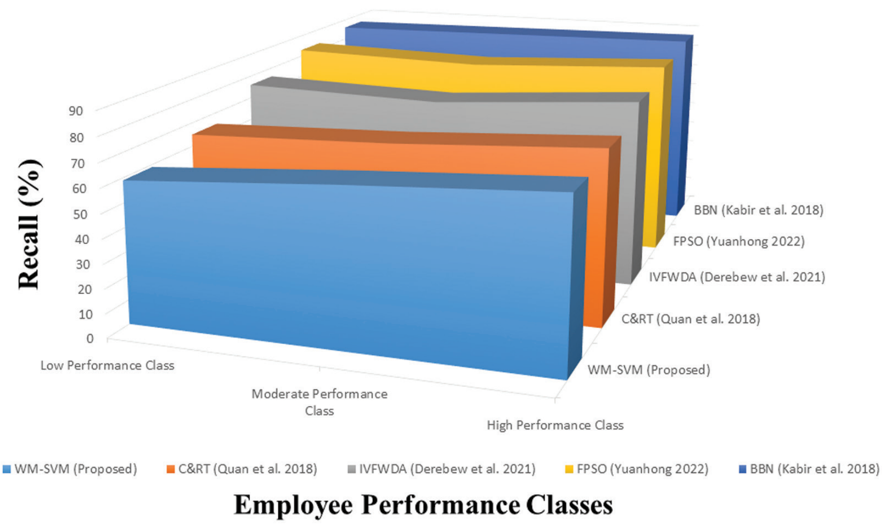

FIGURE 11.7 Comparison of employee performance classification precision of different models.

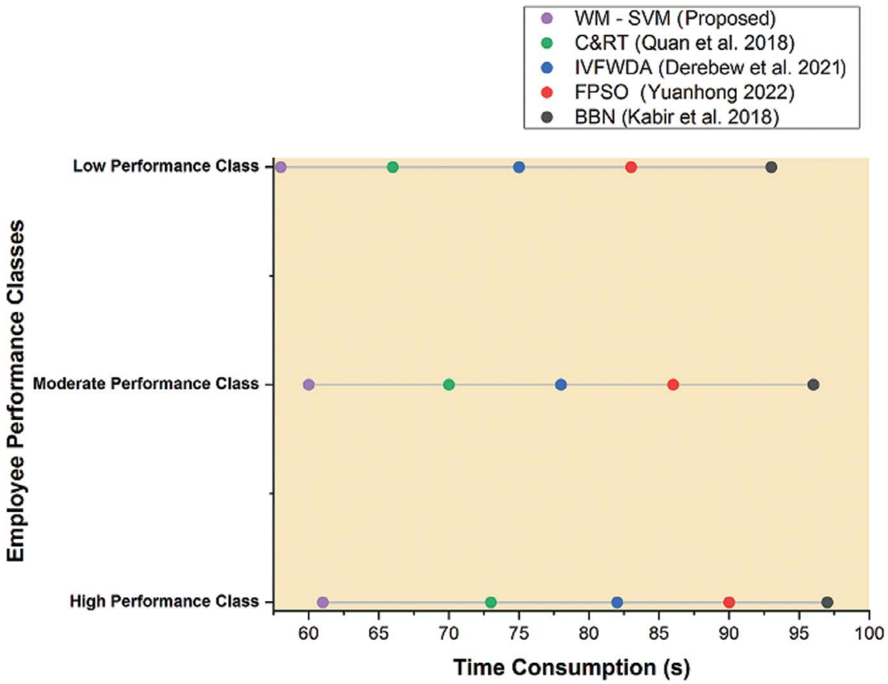

FIGURE 11.8 Comparison of time consumption for employee performance classification using different models.

The proposed WM-SVM–based employee performance evaluation model took less time for employee performance classification compared to existing performance evaluation methods like BBN, FPSO, IVFWDA, and C&RT.

In addition, the time for predicting low-performance employees using WM-SVM was lower than that for predicting moderate- and high-performance employees using WM-SVM. WM-SVM–based employee performance evaluation is a time-efficient model. This model helps the enterprises to evaluate the performance of their employees as shown in Figure 11.8.

Three performance classes considered in this chapter are high performance, moderate performance, and low performance. In this figure, high-performance, moderate-performance, and low-performance classes are denoted by 1, 2, and 3. Forty percent of employees in the firm are predicted to be moderate-performance employees, 35% of employees in the firm are predicted to be high-performance employees, and 25% of employees in the firm are predicted to be low-performance employees.

Figure 11.9 depicts the performance class of each employee predicted by the proposed WM-SVM-based employee performance evaluation model.

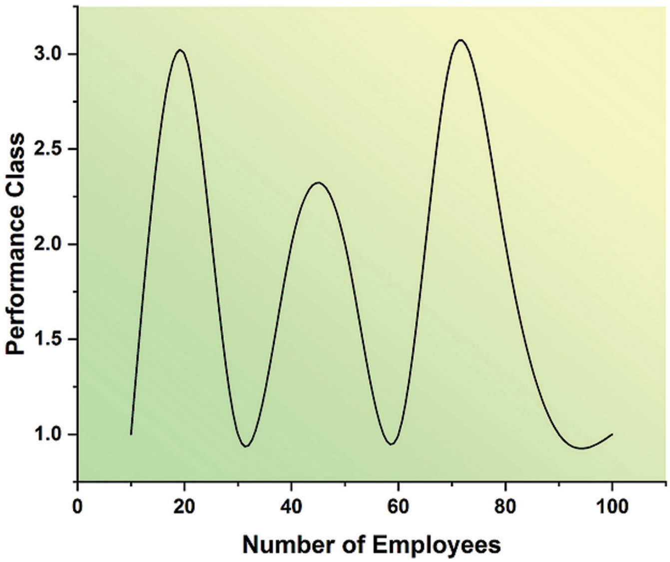

FIGURE 11.9 Number of employees versus their performance class.

11.5 CONCLUSION

An automated employee performance evaluation model is crucially required for identifying the needs for employee and organizational development and motivating and rewarding the employees in enterprises (Hajimahmud et al., 2022).

The main focus of this chapter is to develop an efficient WM-SVM–based employee performance evaluation model to classify the enterprise employees depending on their performance into high-, low-, and moderate-performance groups. The efficacy of the WM-SVM–based employee performance evaluation model was compared to existing performance evaluation methods like BBN, FPSO, IVFWDA, and C&RT.

Accuracy, precision, and recall for classifying the enterprise employees using WM-SVM were higher than that of existing performance evaluation methods like BBN, FPSO, IVFWDA, and C&RT.

Moreover, the WM-SVM–based employee performance evaluation model is time-efficient compared to conventional models like BBN, FPSO, IVFWDA, and C&RT. Hence, the WM-SVM–based employee performance evaluation model can be adopted for employee management in enterprises (Tailor, Pareek & Khang, 2022).

The limitations of this work are that the WM-SVM–based employee performance evaluation model is facing challenges like trust and accountability of employee evaluation data. This means that in the future, the WM-SVM–based employee performance assessment model will need to include technology that guarantees the trustworthiness and transparency of employee evaluation data (Khang et al., 2022c).

In this chapter, an ML-based employee performance evaluation model was developed. In the future, a deep learning-based employee performance evaluation model can be developed to further improve employee performance classification (Hussain, Sivakumar & Khang, 2022).

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12 AI-Enabled Approaches and Models for Designing the Workforce through Training Systems for Physically Challenged People

Babasaheb Jadhav, Ashish Kulkarni,
and Pooja Kulkarni

CONTENTS

12.1 INTRODUCTION

As of now, there is no much platforms that provide an easy way of conducting online classes for disabled students. Using IoT devices can help to improve the quality of teaching, learning, and timely accessibility of online education. We can use different

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types IoT devices, such as AI cameras, ultrasonic sensors, smart bands with pulse oximeter sensors and SpO_2 sensors, and pointing sticks (Khang et al., 2023b).

Previously, disabled students could not attend online classes because there were no such tolls and facilities to fulfill their needs to be tutored by teachers. On this platform, an artificial intelligence camera can be used to capture the teaching, learning an evaluation records of students and teachers. Then, we can convert it to the appropriate formats, which will be referred by everyone while attending lectures on the same platform (Rana et al., 2021).

Also, disabled students can attend online classes with normal students because we are using speech-to-text and text-to-sign conversion technology to help disabled students to understand teachers conversation and vice versa by using AI (Khanh et al., 2021).

We are also using a health monitoring system for disabled students, which will notify their parents, teachers, or hospital. If they face some abnormality in their health condition, then smart bands with pulse oximeter and SpO_2 sensors measure the chronic level. We also use the pointing stick to move the cursor on the screen for students with hand disabilities, so that they can move the cursor on the screen to interact, chat, discuss, and participate in the learning.

In addition, if students want to learn life skills like playing guitar, playing drums, or learning indoor games, the required apparatus should make available virtually for learning (Snehal et al., 2023).

12.2 PROBLEM STATEMENT

Classification of basics of disabilities is as follows.

12.2.1 Visual Impairment

In visual impairment, disabled students use their voice commands, and the system will automatically convert the voice into text using speech-to-text conversion technology to write and communicate with teachers with the help of a built-in mic in the smart band. The built-in AI camera will have a push-to-talk feature and special monitors that capture the faces of students and teachers (Babasaheb et al. 2023a).

12.2.2 Hearing Impairment and Speech Disorder

In hearing impairment, we can use the speech-to-text conversion on the student side and text-to-speech on the teacher side with the help of a keyboard where students can type and communicate. With help of an AI camera, students can use sign language to communicate, which will be converted into text for teachers (Babasaheb et al., 2023b).

Students who cannot speak and want to seek the attention of the teacher in the middle of the lecture, the student can just type, and the system will convert typed text into speech that can be heard by the teacher and other students present in the lecture.

12.2.3 Physical Disability

In a physical disability, there could be any scenario. Depending upon the scenario, we can use different devices to interact with the screen and communicate in online classes. For students with hand disabilities, we can use a pointing-stick device to assist the student, in which the student can move the cursor on the screen and a speech recognition system will help them with it.

12.3 LITERATURE REVIEWED

Svalin and Ivić (2020) discussed enforcement of the pandemic and forceful adoption of online teaching and learning. Authors are concern about the involvement of parents specifically mothers to intervene in online education for a disabled child.

Kent (2016) done his study before the pandemic around 2014–15 in Australia. It is done in two ways: first, the student's experience and second, the student approach. The study is a questionnaire-based approach where the outcome was a discussed on various types of policy-level work and student preference for technology.

Cronin (1996) discussed several programs to address life skills and instructions for students with learning disabilities.

Hendra Jaya (2018) discussed the importance of life skills. Due to these life skills, students will live independently. These modules or courses help them to understand the content of the lessons and imbibe vocational skills.

As there is very little literature available for teaching life skills through online education, this attempt by the authors may help to increase the research and provide education as a birth right for all these special children.

12.3.1 Objectives and Methodology

The primary objective of the approaches and model is to provide an artificial intelligence-based online teaching system that provides easy access to education for disabled students and enables them to actively participate in online classes, thereby enhancing their online education (Gupta et al., 2023).

Another objective is to provide an artificial intelligence-based online teaching system for disabled students that monitors them and helps them to interact with the online class (Khang et al., 2022a).

Yet another objective of the approaches and model is to provide artificial intelligence-based online teaching system that understands sign language and converts that into text.

The other objective is to provide an artificial intelligence-based online teaching system that helps visually impaired students listen to the classes with voice-to-text technology.

Another objective is to offer a module or system that provides access to online classes to disabled students using various IoT devices to aid disabled students of different categories such as students with blindness, deaf, and dumb (Khang et al., 2023b).

A further objective is to provide an artificial intelligence-based online teaching system that utilizes a pointing stick device to help a physically disabled student to move the cursor on the monitor.

Another objective is to provide an artificial intelligence-based online teaching system that monitors the health parameters of students and transmits an alert to family and hospital if any abnormality is detected.

12.3.2 Main Work Section

The present study and models have been created to solve the problems of disabled children and provide learning over the network. It is an objective of the present approaches and model to provide an artificial intelligence-based online teaching system. It provides easy access to education for disabled students and enables them to actively participate in online classes (Jain et al., 2023).

According to a prototypical structure of the approaches and model, Figure 12.1 refers to a proposed block diagram of an artificial intelligence-based online teaching system for disabled students. The artificial intelligence-based online teaching system comprises at least two camera modules, a sign language recognition module, a communication module, a conversion module, and a pointing device (Khang et al., 2022b).

The artificial intelligence-based online teaching system is an application suitable for conducting online classes using various electronic devices, such as a mobile, a laptop, or a computer with a webcam.

The camera modules are configured to recognize and capture the movements of a student and a teacher on their respective devices. One camera module is configured on a teacher's device and the other camera module is configured on a student's device. The sign language recognition module is configured to recognize the sign language (body language) of the student and the teacher using the captured movements of the student and the teacher from the camera modules.

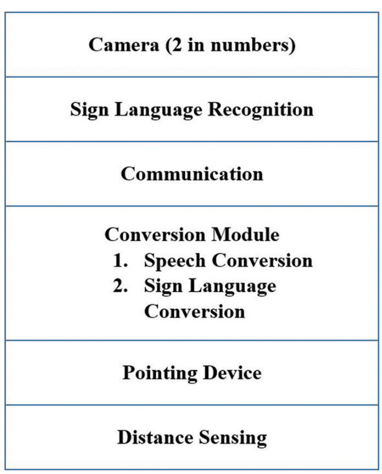

FIGURE 12.1 Block diagram of an artificial intelligence-based online teaching system.

The communication module is configured to establish communication between the teacher and the student by transferring audio to and from the teacher and the student. The conversion module is configured to translate the input language into the required output language based on the input language provided by the teacher and the student. The conversion module is integrated into a wristband that utilizes speech-to-text and text-to-speech converting algorithms for conversion.

The conversion module comprises a speech conversion module and a sign language converting module. The speech conversion module is configured to translate either text to speech or speech to text to enable easy communication between the student and the teacher. After conversion, the converted output is on a display unit.

Furthermore, the pointing device is also integrated with the speech recognition technique to move the cursor using voice commands. Additionally, the system comprises a distance-sensing module that is configured to measure the distance between the students from the display unit (Khang et al., 2022c).

The distance-sensing module comprises an ultrasonic sensor for the measurement of distance, which provides an alert to the student based on the distance measured by the wearable wristband.

The wearable wristband consists of a pulse oximeter sensor and a SpO2 sensor to monitor the health of the student and transmit a notification for any abnormality in the health of the student. The artificial intelligence-based online teaching system monitors the health parameters of every student and transmits an alert to the family and hospital.

The model comprises a microphone that automatically turns on when the student is at a threshold distance from the display unit and turns off if the student is far from the display unit as in Figure 12.2. According to another exemplary embodiment of the approaches and model (Hajimahmud et al., 2022).

At this stage, if, no presence is detected that means, the distance is more than a threshold limit and an alert is sent to the students. Then, once again the distance sensing module measures the distance. Once the student comes within the threshold limit, then camera recognize and capture the movements and faces of the student and the teacher.

If the camera modules recognize the face of the student and the teacher, then system will continue to operate. Otherwise, the system goes back to the previous step and repeats it until the camera modules recognize the faces of the student and the teacher. The system enables the students and teachers to select an appropriate class for teaching and learning. A visually impaired student is aided with text-to-speech conversion functionality.

The visually impaired student can hear the texts typed by the teacher. At the next step, a hearing-impaired student is aided with speech-to-text conversion functionality. The hearing-impaired student can read texts on the display unit while the teacher speaks in the class.

In a further step, a physically disabled student is provided with a pointing device that allows the student to move the cursor on the display unit. Numerous advantages of the present models and approaches may be apparent from the discussion above.

With the present models and approaches, an artificial intelligence-based online teaching system provides easy access to education for disabled students and enables

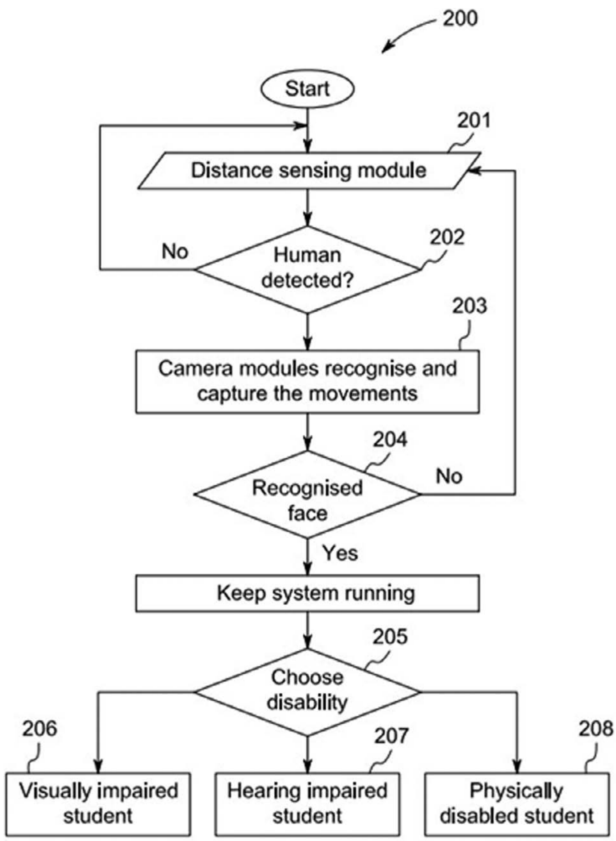

FIGURE 12.2 Working method of the artificial intelligence-based online teaching system.

them to actively participate in online classes, thereby enhancing the online education of disabled students.

The proposed artificial intelligence-based online teaching system for disabled students monitors and helps disabled students to interact with the online class. The system monitors the health parameters of students and transmits an alert to families and hospitals.

The system also converts classes into an appropriate format that is understandable by everyone attending classes on the same platform. It is suitable for conducting online classes using various electronic devices such as smart phones, laptops, or computers with a webcam.

This system provides access to online classes using various Internet of Things (IoT) devices to aid disabled students of different categories (Rani et al., 2021). It will readily be apparent that numerous modifications and alterations can be made to the processes described in the foregoing examples without departing from the principles underlying the approaches and model, and all such modifications and alterations are intended to be embraced by this application (Bhambri et al., 2022).

12.4 THE CASE-BASED APPROACH

The cases where students with disabilities can come forward and do some miracle in the respective field. One of the very common cases known as Mr. Mark Zuckerberg. He is one of the richest persons in the world and he is color blind. This was the reason behind keeping Facebook logo in blue color.

There are many such case studies where people with disabilities create success stories. A few of them are as follows:

12.4.1 Terrasinne

Café available on FC road Pune Maharashtra's, successfully owned and run by deaf, and specially abled person.

12.4.2 RK Fashions

Mr. Ranjeet Kumar from the family of carpenters joined a training Program during a pandemic and started his own business, which now is successfully running in Delhi, Jaipur, Agra, Mumbai, Jammu, and Nagaland.

12.4.3 Nitin Arts

Mr. Nitin Kishan Chand, an entrepreneur with hypothyroidism and general disability, learned to draw and paint in Muskaan Special School, New Delhi before embarking on his business journey.

12.4.4 Social Cipher

Vanessa Gill (Los Angeles, California) created Social Cipher, a social-emotional learning platform offering games and curriculums designed to help neurodiverse youth develop learning skills and construct positive boundaries.

12.4.5 Fetchadates

Sheryl Mattys (Westfield, Indiana) formed Fetchadates, a social networking app for single pet lovers to connect with fellow animal lovers.

There are several cases in which if we provide proper training and some handholding, then the people disabilities will do the miracles in sports, technology, and entrepreneurship.

12.5 CONCLUSION

The present models and approaches propose an artificial intelligence-based online teaching system for disabled students. The researcher presents a simplified summary to provide a basic understanding of some aspects of the claimed subject matter.

This summary is not an extensive overview. It is not intended to identify key/critical elements or to delineate the scope of the claimed subject matter. Its sole purpose is to present some concepts in a simplified form as a prelude to the more detailed description that is presented later (Khang et al., 2022d).

To overcome the above deficiencies of the prior art, the present models and approaches are intended to solve the technical problem to provide an artificial intelligence-based online teaching system. It provides easy access to education for disabled students and enables them to actively participate in online classes, thereby enhancing their online education. As per the above outcomes, the approaches and model provide an artificial intelligence-based online teaching system for disabled students.

The artificial intelligence-based online teaching system comprises at least two cameras, a sign language recognition, a communication module, a conversion module, and a pointing device. It is an application suitable for conducting online classes using various electronic devices such as a mobile phone, a laptop, or a computer with a webcam.

The camera modules are configured to recognize and capture the movements of at least one student and at least one teacher at their respective ends. One camera module is configured on a teacher's device and the other camera module is configured on a student's device.

The sign language recognition is configured to recognize the body language of the student and the teacher using the captured movements from the camera modules. The communication module is configured to establish communication between the teacher and the student by transferring audio to and from the teacher and the student.

The conversion module is configured to translate the input language into the required output language based on the input language provided by the teacher and the student. The conversion module is integrated into a wristband that utilizes speech-to-text and text-to-speech converting algorithms for conversion. The conversion module comprises a speech conversion module and a sign language converting module.

The speech conversion module is configured to translate either text-to-speech or speech-to-text to enable easy communication between the student and the teacher. The student and teacher conversion displayed with converted output on a display unit. The sign language converting module is configured to convert sign language into text form and display it to the teachers and the students on the display unit.

The pointing device is configured to assist the student to move a cursor on the display unit. Furthermore, the system comprises a distance-sensing module that is configured to measure the distance between the students from the display unit.

12.6 THE FUTURE OF WORK

In this chapter, the study has been made on the flowchart and algorithms, which can be used to teach life skills to disabled students (Vrushank, Khang & Rani, 2023).

These skills are required for their growth in terms of living in society as well as daily needs. This system also takes care of their health with an oximeter and other equipment.

The only major barrier is that the system involves more cost and heavy hardware, which needs to be installed at the student's end. To check the operations and functionality, a working internet connection is required (Gupta et al., 2023).

If the device does not work or the internet connection fails, which frequently happens due to power stability, human intervention is required. In future robot can do all the prerequisite tasks.

Researcher thought to reduce the cost by sharing some devices and based on personal analysis as well as checking the feasibility. Not much literature is available to handle such cases, hence it is very difficult the predict outcome of such activity (Khang et al., 2023b). It must be on a trial-and-run basis.

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13 Relevance Analytics of Work Motivation and Job Satisfaction in the Era of Industry 4.0

Mohammad Qais Rezvani, Hayri Uygun, Rashmi Gujrati, Shivangi Singh, and Nirmala Chaudhary

CONTENTS

13.1 INTRODUCTION

The Fourth Industrial Revolution has been in full swing since 2011, and it has spread throughout the workplace since then. It uses electronics and information technology to automate the manufacturing process. There are some fundamental features, such as autonomous robots, big data and analytics, simulation, the cloud, and augmented reality.

The Fourth Industrial Revolution has brought several advantages to the market and the world, including improved product quality, shorter lead times, and improved stakeholder cooperation. Industry 4.0 emphasizes the importance of an organization's ability to adapt and modify its business practices on a daily basis, as well as increase staff capabilities and adjust technology.

Every organization must strike the right balance between themselves and their employees, and make much more efforts in their workers' job motivation so that the workers feel appreciated and fulfilled on the job (Khang et al., 2023a). Employees who are satisfied with their jobs and with the organization as a whole can contribute to the success of both their jobs and the organization.

The main objective of current study is to provide definitions of work motivation and job satisfaction to organizations and experts so that they can gain a better

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understanding and possibly improve performance in their organizations (Babasaheb et al., 2023a).

13.2 RELATED WORK

13.2.1 Industry 4.0

To enhance the maximum profit for organization. The goal of this chapter is to provide definitions of work motivation and job satisfaction to organizations and experts so that they can gain a better understanding and possibly improve performance in their organizations (Babasaheb et al., 2023b).

Rani et al. (2021) defines that Fourth Industrial Revolution that enhances the workplace delivery of technologies and paradigms that social product development, cloud-based manufacturing, and the Internet of Things (IoT).

Industry 4.0 is concerned with the combination of the virtual and physical worlds through Cyber-Physical Systems (CPS) and the IoT via smart objects that constantly communicate and interact with one another to achieve a predetermined strategic goal in an organization (Gupta et al., 2023).

Realizing Industry 4.0 in an organization is a significant strategic decision for improving capacity for profit, and organizations must assess their readiness for such a major decision with their employees before making it.

The traditional model approach is one of the most frequently used techniques for determining the maturity of an organization or process in order to determine its ability to meet the intended goals, as shown in Table 13.1.

The Fourth Industrial Revolution aims to increase effective competence, production, and automation in an organization. It accelerates the automation process, adaptation, customization, digitalization, and production of human-machine interaction (HMI)

TABLE 13.1
Several industry readiness and maturity models

Instrument	Authors	Number of Dimensions	Scale
IMPULS – Industry 4.0 Readiness	Schumacher, Erol and Sihn (2016)	9	5-point Likert scale reaching from 1 (not distinct) to 5 (very distinct)
Readiness of Industry 4.0	Nimawat (2021)	16	5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree)
Readiness of Industry 4.0	Rajbhandari et al. (2022)	7	5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree)
Industry 4.0 Adaptation Potential (4IRAPS)	Sozbilir (2021)	4	5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree)
Industry 4.0	Jayashree et al. (2021)	3	5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree)

(Hajimahmud et al., 2022). There are major features of Industry 4.0, such as value-added services, automatic data exchange, and communication in business (Khang et al., 2023b).

As a result, by reviewing existing literature, this chapter conducts a broad assessment of the Fourth Industrial Revolution and provides an overview of its topic, instruments, findings, and conclusion (Jain et al., 2023).

13.2.2 Work Motivation

The First Industrial Revolution started around 1760 and lasted until 1840. The main goals were to switch from using human muscle to specialized machinery and from one-on-one work to factories and bulk production. The success of the revolution at the time depended heavily on the iron and textile sectors, in addition to the development of the steam engine.

The Second Industrial Revolution had a very different effect on people's lives; it led to an improved banking, transportation, and factory system that was dependent on their holders and managers in an organization.

The Second Industrial Revolution has been characterized by a greater emphasis on human factors. This was due to an increase in the number of employees who were in mass production, which resulted in more workplace problems.

Furthermore, division of workers must break down job activities into smaller chunks in order to maximize proficiency and deal with a large number of employees. As a result, employee behavior is becoming more important in terms of business productivity.

The world reached the Third Industrial Revolution, which was a milestone at the time of the second-highest peak in human behavior studies. Workers' monotonous behavior was highlighted by organization owners, and their activities were organized into specialized types of mechanical operations (Sonawane et al., 2023).

The shift and transformation from the Third Industrial Revolution to the Fourth Industrial Revolution occurred through a massive psychological shift among industry employees in an organization. The demand for skills and expertise is becoming more diverse and unique than ever before.

Furthermore, understanding work motivation is as important as understanding the other challenges outlined in Industry 4.0 for labor. Motivation is defined as the process of changing and directing human actions into desired work patterns in a workplace, which can include activating behavior, maintaining behavior over time, improving performance standards on specific duties, or all of the above (Aggarwal et al., 2023).

Human psychological processes that lead, excite, and maintain activities toward job-related behavior are called work motivation. Many academics believe that technological solutions will never fully replace humans in the workplace, as shown in Table 13.2. Despite the fact that work motivation is an important topic in organizational behavior, few work motivation questionnaires are available.

13.2.3 Job Satisfaction

Technological advancements in the workplace have altered the way professionals work, causing a shift in their work-life balance toward a greater emphasis on

TABLE 13.2
Work Motivation

Instrument	Authors	Number of Dimensions	Scale
Assessment of Individual Motives –Questionnaire (AIM-Q)	Bernard et al. (2008)	5	4-point Likert scale, ranging from 1 (strongly disagree), 2 (disagree), 3 (agree), to 4 (strongly agree)
The Situational Motivation Scale (SIMS)	Guay, Vallerand and Blanchard (2000)	4	7-point Likert scale ranging from 1 (does not correspond at all) to 7 (corresponds exactly)
A Worker Motivation Scale	Wherry and South (1977)	14	5-point Likert scale of applicability from 1 ("I almost never feel that way") to 5 ("I almost always feel that way")
Work Extrinsic and Intrinsic Motivation Scale	Tremblay et al. (2009)	18	5-point Likert scale ranging from 1 (does not correspond at all) to 5 (corresponds exactly)

Table 13.2 depicts the instruments used for measuring work motivation are as above.

their personal lives without jeopardizing organizational commitments. Employees can accomplish these features through Industry 4.0, which improves connectivity (Khang et al., 2022a).

They can use video conferencing software such as Skype and Zoom to hold meetings with clients and colleagues in other countries. These landscapes reduce the frequency of travel for workers, allowing them to spend less time away from their homes, families, and friends.

There is widespread concern about job losses since robots will become more involved in factories in the future. Robots will not replace humans in the workplace but will have a positive influence on the workplace and work-life balance.

The technology which is introduced by Industry 4.0 has changed the routine activities of workers in all organizations. It may affect shifting the balance between work and personal lives and contribute to the realization of Obama's thought and vision.

The Fourth Industrial Revolution is causing dramatic changes in the work life of employees, such as rapid change in technology, new job profiles with new job descriptions, or simply the ability to connect and collaborate with one another in a smart environment using smart devices.

Employees are more likely to be happy with some facets of their work or particular training models but less content with others. Perhaps their organization does not provide sufficient rewards for their performance, or their workplace is not a suitable fit for them.

To decide what is best for the organization and the employees in the workplace, managers must become familiar with their staff's preferences.

Authors define job satisfaction with their own perspectives and definitions. Job satisfaction is defined by whether employees feel satisfied about their jobs and the various features of those jobs. It can define, in other words, job happiness by determining and measuring of how much people like or dislike their jobs (Shah et al., 2024).

Job satisfaction is often defined to different degrees and can be studied from a variety of outlooks using a variety of tools and classifications. This will be covered later in a discussion of measurements that show how diverse certain categories can be.

A person, for example, may be pleased with some particular parts of his or her job while dissatisfied with others. Job satisfaction is defined as how people feel about their jobs and various aspects of those jobs (Rana et al., 2021). Here are numerous persuasive reasons why companies should be concerned with employee satisfaction. Above all, humans deserve to be treated fairly and with dignity in an organization without regard to gender, race, language, etc.

To some extent, job satisfaction reflects good treatment in the industry. Everyone wants to be treated fairly and equally at the same time and the same level, so working on those features for the benefit of their employees appears to be a good place to start for any company. This viewpoint also contends that external factors influence job satisfaction.

However, there is an internal factor that influences how one feels. Job satisfaction can be defined as a set of characteristics when combined properly in a sense of fulfilment. In addition, Vroom emphasizes job satisfaction in an unusual way. His attention is drawn to the employee's position in the workplace when it comes to job satisfaction. In this case, job satisfaction could be defined as an individual's emotional reaction to various work roles within an organization.

Job satisfaction is defined by Khang et al. (2024) as a collection of attitudes and perspectives about one's current job in an organization. Depending on the business, a person's level of satisfaction might vary, just like their position in an organization. They can experience extremes of happiness or unhappiness, but they still have desires and needs for contentment.

Employee attitudes toward their jobs vary both generally and specifically. For instance, the type of work they do; their relationships with coworkers, bosses, or subordinates; and their income are all factors that have an effect on the job satisfaction of an employee in an organization.

Furthermore, Kaliski (2007) defines job satisfaction as having a sense of success and accomplishment toward the job. Simultaneously, it is thought that job satisfaction is related to productivity, efficacy, and inner happiness.

This section can also refer to job satisfaction, which leads to recognition through higher pay, promotion, and goal achievement, which contribute to a sense of accomplishment in the workplace both for the employees and employers.

13.3 RESULTS AND DISCUSSION

This chapter reviews and evaluates inclusive and a comprehensive on Industry 4.0 with considering and existing relevance analytics and providing an overview of Industry 4.0's in content, scope, and conclusions of the chapter. For instance, industrial revolutions are complex and adaptable processes that involve efforts of both humans and machines in an organization, as shown in Table 13.3.

TABLE 13.3
Job Satisfaction Measurement

Instrument	Authors	Number of Dimensions	Scale
Job Satisfaction Survey (JSS)	Spector (1985)	9	6-point Likert scale ranging from 1 to 6
Minnesota Satisfaction Questionnaire (MSQ)	Weiss (1967)	20	6-point Likert scale ranging from 1 to 6
McCloskey/Mueller Satisfaction Scale (MMSS)	Mueller and McCloskey (1990)	8	5-point Likert scale ranging from 1 to 5
Measure of Job Satisfaction	Traynor et al. (1993)	5	5-point Likert scale ranging from 1 to 5
The Job Training and Job Satisfaction Survey	Schmidt (2004)	2	6-point Likert scale ranging from 1 to 6

Table 13.3 depicts the most frequently used instruments for job satisfaction measurement.

Industry 4.0 reduces costs and increases productivity, at the same time it improves the product and has positive impact on its quality. However, there is still a long way to go beyond production to be able to match all theories in every way (Tailor, Pareek & Khang, 2022).

Work motivation studies investigate the underlying causes of human behavior in the Fourth Industrial Revolution. It causes the fourth manufacturing revolution ushers in a different way. When the task demands change in an organization in the same time automatically the critical work inputs and outputs lists are updated and changed.

As these factors are significant to current organizational accomplishments, more interactive data analysis should be used to investigate key work necessities and their relevant features (Khanget al., 2022a).

13.4 CONCLUSION

Research reveals that current work motivational models will be extremely useful to organizations in developing new job happiness, facilitating employee recruitment, retraining, and redesigning work (Rajah & Tan, 2019).

One of the most difficult tasks for today's managers is how to handle organizational commitment and job satisfaction in an organization. Numerous studies have shown that job satisfaction has an affirmative and massive effect on work motivation, increasing productivity and performance appraisal.

Furthermore, work motivation has a significant impact on job satisfaction, leading in any positive sentiments that convoy a human who seeks to maintain this state for as long as possible, making to additional efforts in the Industry 4.0 scenario (Khang et al., 2022c).

As a result, Industry 4.0 appears to be a vision for the future, as it covers a wide scope in the workplace. Finally, organizations may face a variety of challenges if work motivation and job satisfaction in industry 4.0 is not considerable that may cause economically, scientifically, technologically, socially, and politically issues.

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